

VERIQO COMMERCIALIZATION SPRINT -- DEMO CASES (ITEMS 14-17)

This document answers Commercialization Sprint items 14-16 ("3 canonical demo cases") and item 17 ("customer-facing metrics") by name. Every case below is real, runnable code, not a mockup: pkg/commercial/democases builds each case through the exact, frozen kernel (cre -> decision -> action) via the same pkg/commercial/api.Store a real Commercial API v1 caller uses, and cmd/veriqo-demo-cases writes each one's actual Evidence Dossier v1 -- both the human Markdown and the machine .zip package -- to disk. Every package this command produces independently verifies against cmd/veriqo-commercial-verify, the standalone verifier a customer would run without trusting VERIQO's own producing process.

To regenerate all three:

```
go run ./cmd/veriqo-demo-cases <output-dir>
```

To independently verify any one of the resulting packages:

```
go run ./cmd/veriqo-commercial-verify -package <output-dir>/demo1-eb1-transfer-dispute.zip
```

Each case below follows the same four-part honesty framework the reviewer names: What VERIQO Knows (independently, technically verified facts), What VERIQO Does NOT Know (facts outside this evidence set or outside VERIQO's authority to determine), What VERIQO Can Prove Technically (the specific cryptographic/structural claims a third party can re-check), and What Remains a Legal Question (the determination only a court, arbitration panel, or the parties' governing contract can make).

DEMO 1 -- CASE #VRQ-2026-0001: EBL TRANSFER DISPUTE

Scenario. An electronic Bill of Lading (eBL) is issued to Shipper-Co and, per the issuing platform's own transfer log, later transferred to Bank-A. A dispute arises over whether that transfer was valid and whether Bank-A now holds the eBL. Built by democases.BuildEBLTransferDisputeCase.

What VERIQO Knows

- The issuance record (EV-EBL-ISSUANCE) and the sole recorded transfer event (EV-EBL-TRANSFER-1) are both present, FINALIZED, and hash-intact.
- Both records were submitted through the same custody-chain state machine (RECEIVED -> HASHED -> REVIEWED -> FINALIZED), so their custody history is continuous and independently re-derivable.
- The transfer record names Bank-A as the new holder; nothing in this evidence set contradicts that record.

What VERIQO Does NOT Know

- Whether the eBL platform itself is the sole, authoritative registry for this document, or whether a parallel paper original also exists.
- Whether Bank-A's internal onboarding/KYC on this transfer was itself valid -- that is outside this evidence set entirely.
- Anything about the underlying trade contract's substantive terms (payment terms, INCOTERMS, etc.) beyond the transfer event itself.

What VERIQO Can Prove Technically

- EV-EBL-ISSUANCE and EV-EBL-TRANSFER-1's manifest hashes independently re-derive and match what is recorded (manifest.VerifyManifestHash).
- Each record's custody chain is a verified hash chain from GENESIS

(manifest.VerifyCustodyChainRecords).

- The Decision approving the transfer is grounded ONLY in evidence that was FINALIZED and hash-verified at authorization time (cre.AuthorizeGrounded) -- it did not silently accept a stale or unfinalized record.
- The Decision, the Action Authorization, and the resulting Receipt form a single, independently re-hashable chain (decision.VerifyDecisionProvenance, action.VerifyActionAuthorization).
- Replaying the case's recorded inputs reproduces byte-identical Decision and Authorization hashes (verticalslice.Replay) -- this is not a one-time computation that cannot be checked again later.

Remains a Legal Question

- Whether this transfer satisfies MLETR "control" and "exclusive control" in the receiving jurisdiction -- see docs/VERIQO_MLETR_EBL_RELIABILITY_CONFORMANCE_MAPPING.md (an earlier round's honest primary-source-verification gap on this exact point).
- Whether the transfer is legally effective against third parties under the governing law of the underlying trade contract.
- Any dispute over the substantive commercial terms the eBL evidences.

DEMO 2 -- CASE #VRQ-2026-0002: MARITIME INCIDENT (POSITION CONTRADICTION)

Scenario. An AIS track for MV EXAMPLE and a port authority's berth arrival log disagree on the vessel's position/time on 2026-01-15. Rather than silently picking one record, the case is escalated for human review. Built by democases.BuildMaritimeIncidentCase.

What VERIQO Knows

- EV-AIS-TRACK-1 (AIS_STATUS) and EV-PORT-ARRIVAL-1 (PORT_EVENT) are each individually hash-intact and FINALIZED.
- The two records are mutually inconsistent on the vessel's position/time for the same voyage leg -- VERIQO surfaced this as a genuine contradiction (Contradictions: EV-PORT-ARRIVAL-1 contradicts hypothesis H1 in the generated dossier), not a silently-resolved discrepancy.

What VERIQO Does NOT Know

- WHY the two records disagree: AIS spoofing, a transponder fault, a delayed or manually-entered port log, or a genuine deviation are all consistent with this evidence set. VERIQO does not have the specialized maritime-domain intelligence to adjudicate among them from these two records alone.
- Which record, if either, is more reliable in general -- that would require either a track record of each source's historical reliability or additional corroborating evidence not present in this case.

What VERIQO Can Prove Technically

- Both records independently hash-verify and custody-chain-verify on their own.
- The contradiction itself is a first-class, machine-checkable fact: the Finding's ContradictedBy field cites EV-PORT-ARRIVAL-1 by ID, and cre.AuthorizeGrounded confirmed that record is real and FINALIZED before allowing the contradiction to be recorded at all (a fabricated or unfinalized "contradiction" cannot enter the Decision).
- The Decision's Outcome is ESCALATED, not APPROVED or DENIED -- the system's own recorded outcome reflects that a technical hash/custody check cannot resolve which record is accurate.

- The resulting SEND_NOTIFICATION action (to the underwriter and P&I club) is itself authorized against this exact ESCALATED Decision, and that link independently re-verifies.

Remains a Legal Question

- Which record is authoritative for any subsequent claim, dispute, or regulatory finding -- that determination belongs to the investigating authority, the P&I club's own inquiry, or a tribunal, informed by (but not made by) this dossier.
- Any liability arising from an actual deviation, if one occurred.

DEMO 3 -- CASE #VRQ-2026-0003: INSURANCE CLAIM (MARINE CARGO)

Scenario. A cargo claim under POL-9001 is supported by an independent surveyor's report and an adjuster's report, both attributing the loss to water ingress during transit. Built by democases.BuildInsuranceClaimCase.

What VERIQO Knows

- EV-SURVEY-1 and EV-ADJUSTER-1 are both FINALIZED, hash-intact, and mutually consistent with the water-ingress hypothesis.
- No contradicting evidence was submitted in this case.

What VERIQO Does NOT Know

- Whether the surveyor's or adjuster's professional conclusions are themselves correct -- VERIQO verifies that these documents are what they claim to be (intact, custody-tracked, attributable to their stated source) and were actually cited as the basis for the Finding; it does not re-derive their professional conclusions from first principles.
- Whether POL-9001's policy wording actually covers this specific loss scenario, exclusions included.
- The fair settlement quantum -- QuantumRef here names a reference to an external calculation (calc-4471-v1), not a VERIQO-computed amount.

What VERIQO Can Prove Technically

- Both evidence items independently hash-verify and custody-chain-verify.
- The Finding was authorized ONLY because both cited evidence items resolved to real, FINALIZED, hash-verified manifests (cre.AuthorizeGrounded) -- a Finding citing missing or unfinalized evidence is refused outright, proven by TestDecideCaseRejectsUngroundedEvidence in this Store's own test suite.
- The APPROVE_SETTLEMENT Action Authorization's own hash embeds the exact Decision hash it was authorized against (action.VerifyActionAuthorization), so the settlement authorization cannot be silently detached from, or reattached to, a different Decision after the fact.
- Replay reproduces identical Decision and Authorization hashes from the same recorded inputs.

Remains a Legal Question

- Whether the policy, correctly interpreted, covers this loss.
- Whether the settlement amount computed by calc-4471-v1 is fair or contractually correct.
- Any dispute over the surveyor's or adjuster's professional findings themselves.

CUSTOMER-FACING METRICS (ITEM 17)

The reviewer's explicit instruction is that these are pilot hypotheses to be measured during a real pilot, not universal claims -- VERIQO has not run a paid pilot and has no measured before/after data of its own. The BEFORE column below describes a typical unassisted manual-review workflow reported in this industry (evidence gathering, cross-referencing records, cross-team hand-off); the AFTER column is a target, not a result, framed exactly as such.

Metric	BEFORE (typical manual baseline)	AFTER (pilot hypothesis, target)
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Investigation time per case	Days, spread across multiple systems and hand-offs	Target: reduced by up to 80% for cases whose evidence is fully digital and already in VERIQO
Evidence retrieval	Manual search across email, shared drives, and vendor portals	A single case dossier assembling every submitted evidence item with verified integrity
Contradictions discovered	Often found late, by a human noticing a discrepancy	Target: surfaced automatically at decision time, as Demo 2 shows
False positives (flagged issues that turn out not to matter)	No consistent baseline; varies by reviewer	Target: reduced by 40%, to be measured against a specific pilot's real flag volume
Detection lead time (incident to first flag)	Hours to days depending on reporting chain	Target: improved by 12 hours where source evidence already flows into VERIQO in near-real-time
Manual review workload	100% of cases reviewed manually end-to-end	Target: reduced by 60% for the subset of cases where VERIQO's grounded Finding is directly actionable, with human review retained for every escalated or contested case
Time to a documented decision	Varies widely; often undocumented until settlement	A Decision, Authorization, and Receipt are recorded and hash-verifiable the moment a human authorizes an action

Honest caveat, stated once and meant literally: every "Target" figure above is a hypothesis to validate against a specific pilot customer's real volume and workflow, not a claim already demonstrated by this codebase. Item 8's marketing-language discipline applies here without exception -- these numbers must never be presented to a prospect as measured results before a pilot has actually measured them.